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Quantum transition state theory: Perturbation expansion

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The exact quantum expression for the thermal rate of reaction is the trace of a product of two operators. It may therefore be written exactly as a phase space integral over the Wigner phase space representations of the two operators. The two are a projection operator onto the product's space, which is difficult to compute, and the symmetrized thermal flux operator, which can be computed using Monte Carlo methods. A quantum transition state theory was presented recently, in which the exact projection operator was replaced by its parabolic barrier limit. Alternatively, the exact projection operator may be replaced by its classical limit. Both approximations give thermodynamic estimates for the quantum rates. In this paper, we derive a perturbation theory expansion for the projection operator about the parabolic barrier limit and the classical limit. The correction terms are then used to evaluate the leading order corrections to the rate estimates based on the parabolic barrier or classical limits of the projection operator. The expansion is applied to a symmetric and an asymmetric Eckart barrier. The first two terms in the expansion give excellent results for temperatures above the crossover between quantum tunneling and thermal activation. For deep tunneling and asymmetric systems, the use of variational transition state theory, the classical limit, and perturbation theory leads to significant improvement in the estimate of the tunneling rate. Multidimensional extensions are presented and discussed. © 1998 American Institute of Physics. [S0021-9606(98)01223-9]

I. INTRODUCTION

One of the outstanding problems of the twentieth century in the quantum theory of reactions was the formulation of a quantum transition state theory (QTST). Classical transition state theory was already formally formulated by 1940 through the pioneering work of Eyring¹ and Wigner.² The main assumption of the theory is that any trajectory with positive velocity perpendicular to a dividing surface will be reactive. This assumption allows one to replace the dynamical computation of a thermal rate with a statistical mechanical computation of partition functions.

The classical estimate is an upper bound to the rate, whose quality depends on the choice of the dividing surface. Variational TST (VTST) is obtained by varying the dividing surface so as to minimize the rate estimate. The errors in the classical TST assumption are also well understood. Classical VTST has turned out to be an important practical tool in the estimation of reaction rates for a very large variety of processes as summarized in various reviews on the subject.³⁻⁷ The development of fast computers makes classical VTST especially attractive, even for systems with thousands of particles, it is today possible to estimate reaction rates with this theory.⁸

It is thus not surprising, that parallel to the early developments of the classical theory, attempts were made to formulate a quantum mechanical analog. There is, though, one

fundamental problem. In quantum mechanics, one cannot know the sign of the momentum at a given position on the dividing surface: this would violate the uncertainty principle. This seemingly insurmountable problem did not discourage the community. Wigner, in 1932,⁹ suggested an appealing quantum approximation to the rate. He used the classical TST rate expression but substituted the classical density in phase space with the Wigner density in phase space.¹⁰ His expression is exact for a parabolic barrier potential. It is, however, an ansatz, a remarkably insightful guess, but the quality of the estimate remains unknown, since it was not derived directly from the exact quantum expression for the rate.

Since Wigner, numerous other attempts have been made to formulate a quantum TST.¹¹⁻²⁴ These include separable approximations,¹¹ rigorous bounds (of poor quality),^{12,14-17} semiclassical methods,¹⁸⁻²⁰ and notably the centroid method,²¹⁻²⁴ which has become very popular during the last decade. In the centroid method, one computes a quantum potential of mean force, obtained from the quantum centroid density. The potential of mean force is then used as in the classical theory, to obtain an estimate for the rate. The main advantage of this approach is that the computation of the quantum centroid density demands only diagonal matrix elements, which are readily estimated by Monte Carlo methods, even for large systems. The method is exact for a parabolic barrier. Just as the early Wigner ansatz, the centroid method is an intelligent guess; it is not derived directly from the exact quantum mechanical expression. It is, therefore, not

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a controllable approximation: one does not have a good handle as to the error when applied to a given system.

We have recently proposed a new quantum TST²⁵ which is an extension of an earlier idea, proposed by Voth, Chandler, and Miller.²⁶ The quantum TST is derived from the well known flux-side expression for the quantum rate $k(T)$ at temperature T ,²⁷

$$k(T) = Q_r(T)^{-1} \lim_{t \rightarrow \infty} C_{fs}(t), \quad (1.1)$$

where $Q_r(T)$ is the partition function of reactants and the flux-side correlation function is

$$C_{fs}(t) = \text{Tr}[\hat{F}(\beta, q_{ds}) e^{i\hat{H}t/\hbar} \hat{h}(\hat{q} - q^\ddagger) e^{-i\hat{H}t/\hbar}]. \quad (1.2)$$

$\hat{h}$ is the step function operator which is unity on the product side of a dividing surface and is zero on the reactant side, and $\hat{H}$ is the Hamiltonian operator. $\hat{F}(\beta, q_{ds})$ is the symmetrized quantum thermal flux operator at the dividing surface defined by q_{ds} , which takes the form

$$\hat{F}(\beta, q_{ds}) = e^{-\beta\hat{H}/2} \hat{F}(q_{ds}) e^{-\beta\hat{H}/2}, \quad (1.3)$$

where $\beta = (1/k_B T)$. The symmetrized flux operator $\hat{F}(q_{ds})$ (in one dimension) is

$$\hat{F}(q_{ds}) = \frac{1}{2m} [\hat{p} \delta(\hat{q} - q_{ds}) + \delta(\hat{q} - q_{ds}) \hat{p}]. \quad (1.4)$$

One notes that the exact quantum expression involves a trace of a product of two operators, the symmetrized thermal flux operator and a projection operator,

$$\hat{P} \equiv \lim_{t \rightarrow \infty} \hat{P}(t) \equiv \lim_{t \rightarrow \infty} [e^{i\hat{H}t/\hbar} \hat{h}(\hat{q} - q^\ddagger) e^{-i\hat{H}t/\hbar}]. \quad (1.5)$$

The symmetrized thermal flux operator may be expressed in terms of matrix elements of the thermal density matrix; the major difficulty is the evaluation of the projection operator $\hat{P}$, which in principle involves an infinite time propagation.

Since the exact rate expression is the trace of a product of two operators only, it can be recast exactly as the phase space trace over the Wigner phase space representations of the two operators, defined for any operator $\hat{O}$ as

$$\begin{aligned} \rho_W(\hat{O}; p, q) &= \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} d\xi e^{ip\xi/\hbar} \\ &\times \left\langle q - \frac{1}{2} \xi | \hat{O} | q + \frac{1}{2} \xi \right\rangle. \end{aligned} \quad (1.6)$$

The Wigner phase space representation of the symmetrized thermal flux operator still involves only matrix elements of the canonical density $e^{-\beta\hat{H}}$.

Quantum TST, as suggested in Ref. 25, is obtained by replacing the exact projection operator with its parabolic barrier approximation. The Hamiltonian can always be divided into a parabolic barrier contribution $\hat{H}_{pb}$ and a “nonlinearity” $\hat{H}_1$. The parabolic barrier Hamiltonian takes the form

$$\hat{H}_{pb} = \frac{\hat{p}^2}{2m} - \frac{1}{2} m \omega^2 \hat{q}^2, \quad (1.7)$$

where m is the mass of the reacting particle, ω is the barrier frequency, and $\hat{p}$, $\hat{q}$ are the momentum and coordinate operators, respectively. We assumed without loss of generality that the barrier separating reactants and products is located at $q = 0$.

The Wigner representation of the parabolic barrier projection operator, denoted as $P_{00}^W(p, q)$ (cf. Sec. III, below), is extremely simple,^{25,28}

$$P_{00}^W(p, q) = \frac{1}{2\pi\hbar} h(p + m\omega q). \quad (1.8)$$

Substituting this approximate projection operator into the rate expression gives a thermodynamic estimate for the rate. Contrary to all other previous attempts at formulating a quantum TST, this new approximation is just a leading order term in the expansion of the quantum projection operator, where the nonlinearity $\hat{H}_1$ is the small parameter.

The central topic of the present paper is the derivation of the correction terms to the parabolic barrier projection operator. We will show that the higher order correction terms are readily evaluated, and that they too involve only matrix elements of the thermal density operator $e^{-\beta\hat{H}}$. This turns the new quantum TST into a controllable theory, whose accuracy may be estimated by evaluation of the next order terms in an expansion of the exact projection operator. The expansion is obtained also for multidimensional systems.

As is well known, the classical limit of the Wigner phase space representation of a quantum operator is just the classical limit of the operator. One may thus replace the quantum projection operator by its classical limit.^{25,29} We will refer to this approximation to the rate as the mixed quantum-classical expression to distinguish it from QTST, in which one uses the parabolic barrier projection operator. One can also obtain a series expansion of the quantum projection operator about its classical limit, where the small parameter is $\hbar^2$. The first few terms of this series will also be derived in this paper. For one-dimensional systems, the classical projection operator is simply the Heaviside function located on the classical separatrix,^{25,29} and so the rate estimate remains a phase space integration involving matrix elements of the thermal density operator. We shall also see that this same structure remains when deriving the leading order correction terms. In multidimensional systems, the classical separatrix becomes in principle a complicated function, so that finding the correction terms to the classical limit may become quite difficult, although the evaluation of the classical limit itself may be routinely estimated through the use of classical trajectories.

In Sec. II we present the central recursion relation needed to derive the perturbation expansion. The leading order terms in the series for one-dimensional systems are presented in Sec. III and applied in Sec. IV to the symmetric and asymmetric Eckart barrier. The multidimensional perturbation expansion is considered in Sec. V, and we end in Sec. VI with a discussion of the method, its limitations, and extensions.

II. QUANTUM RECURSION RELATION

A. The recursion relation in configuration space

The projection operator can be recast exactly²⁷ as an infinite time integral over a time-evolved flux operator at the barrier top ($q=0$),

$$\hat{F}(t) \equiv e^{i\hat{H}t/\hbar} \hat{F}(0) e^{-i\hat{H}t/\hbar}, \quad (2.1)$$

where for brevity we suppress the dependence on β and q_{ds} . The time evolved flux operator is known analytically for a parabolic barrier. As mentioned in the Introduction, the full Hamiltonian may be divided into a parabolic barrier Hamiltonian and a nonlinearity [cf. Eq. (1.7)]. In this subsection we derive a perturbation expansion for the time-evolved flux operator in terms of the nonlinearity $\hat{H}_1$.

For this purpose we consider the Liouville operator $\hat{L}$ defined as

$$\hat{L}\hat{A} = [\hat{H}, \hat{A}]/\hbar, \quad (2.2)$$

for any operator $\hat{A}$.³⁰ $\hat{L}$ is a superoperator acting on the operators in the Hilbert space. The set of all ordinary operators forms a linear space called the Liouville or dual space in which the (sup) vector is the operator in the Hilbert space. In the Liouville space the inner product of two operators $\hat{A}$ and $\hat{B}$ is defined as $\text{Tr}(\hat{A}\hat{B})$. One can prove that the superoperator $\hat{L}$ is Hermitian. Similar to the traditional time development operator with respect to a Hamiltonian, the propagator associated with $\hat{L}$ reads

$$\mathcal{U}(t, t') \hat{P}(t') = \begin{cases} \hat{P}(t) & t \geq t' \\ 0 & t < t' \end{cases}, \quad (2.3)$$

with the boundary condition $\lim_{t \rightarrow t'} \mathcal{U}(t, t') = 1$.

The time-evolved flux operator may be thus written as

$$\hat{F}(t) = \mathcal{U}(t, 0) \hat{F}(0) = e^{i\hat{L}t} h(t) \hat{F}(0), \quad (2.4)$$

where $h(t)$ is the Heaviside unit step function.

To work out a perturbation series for $\hat{F}(t)$ we divide the Liouville operator into two parts. One is associated with the parabolic barrier Hamiltonian ($\hat{L}_{pb}$) and the other ($\hat{L}_1$) is associated with the nonlinearity $\hat{H}_1$, such that $\hat{L} = \hat{L}_{pb} + \hat{L}_1$. Using the notation

$$e^{i\hat{L}t} = e^{i\hat{L}_{pb}t} \mathcal{K}(t), \quad (2.5)$$

we find that $\mathcal{K}(t)$ satisfies the equation

$$\frac{d\mathcal{K}(t)}{dt} = i\tilde{L}_1(t) \mathcal{K}(t), \quad (2.6)$$

where $\tilde{L}_1(t) = e^{-i\hat{L}_{pb}t} \hat{L}_1 e^{i\hat{L}_{pb}t}$ is the perturbation term in the “interaction” representation.

Noting that $\mathcal{K}(0) = 1$, we obtain by iteration,

$$\begin{aligned} \mathcal{K}(t) = & 1 + i \int_0^t dt_1 \tilde{L}_1(t_1) \\ & + (i)^2 \int_0^t dt_1 \int_0^{t_1} dt_2 \tilde{L}_1(t_1) \tilde{L}_1(t_2) + \cdots \equiv \sum_{n=0}^{\infty} \mathcal{K}_n. \end{aligned} \quad (2.7)$$

The recursion relation between $\mathcal{K}_n(t)$ and $\mathcal{K}_{n-1}(t)$ reads

$$\mathcal{K}_n(t) = i \int_0^t dt_1 \tilde{L}_1(t_1) \mathcal{K}_{n-1}(t_1), \quad (2.8)$$

so that the time-evolved flux operator may be expanded as

$$\hat{F}(t) = e^{i\hat{L}_{pb}t} \mathcal{K}(t) \hat{F}(0) = \sum_{n=0}^{\infty} e^{i\hat{L}_{pb}t} \mathcal{K}_n(t) \hat{F}(0). \quad (2.9)$$

Thus far, the formal expansion was general. We now treat explicitly the one-dimensional case; the straightforward generalization to multidimensional systems will be given in Sec. V. The configuration space representation of the time-evolved flux takes the form

$$\begin{aligned} F(t, x, y) & \equiv \langle x | \hat{F}(t) | y \rangle = \sum_{n=0}^{\infty} \langle x | e^{i\hat{L}_{pb}t} \mathcal{K}_n(t) \hat{F} | y \rangle \\ & \equiv \sum_{n=0}^{\infty} F_n(t, x, y). \end{aligned} \quad (2.10)$$

From the definition of the Liouville superoperator, one obtains

$$\begin{aligned} \langle y | f(\hat{L}) \hat{A} | x \rangle & = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dx_1 dx_2 \langle y | f(\hat{L}) | x_1 x_2 \rangle \\ & \quad \times \langle x_1 | \hat{A} | x_2 \rangle, \end{aligned} \quad (2.11)$$

where the matrix element of the superoperator is determined in terms of the matrix elements of the Hamiltonian operator as

$$\begin{aligned} \langle x_1 x_2 | \hat{L} | x_3 x_4 \rangle & = \frac{1}{\hbar} [\delta(x_2 - x_4) \langle x_1 | \hat{H} | x_3 \rangle \\ & \quad - \delta(x_1 - x_3) \langle x_4 | \hat{H} | x_2 \rangle]. \end{aligned} \quad (2.12)$$

Note that $|x_1 x_2\rangle$ is a vector in the Liouville space. The closure property reads

$$\mathbf{1} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dx_1 dx_2 |x_1 x_2\rangle \langle x_1 x_2|. \quad (2.13)$$

From the above rules one can derive the useful identity,

$$\langle x_1 x_2 | e^{i\hat{L}t} | x_3 x_4 \rangle = \langle x_1 | e^{i\hat{H}t/\hbar} | x_3 \rangle \langle x_4 | e^{-i\hat{H}t/\hbar} | x_2 \rangle. \quad (2.14)$$

With these preliminaries, using the definition of the symmetrized flux operator Eq. (1.4), we can write the configuration space representation of the time-evolved flux operator as

$$F(t, x, y) = -\frac{i\hbar}{2m} \frac{\partial}{\partial q} [(xy | e^{i\hat{L}t} | 0q) - (xy | e^{i\hat{L}t} | q0)]_{q=0}. \quad (2.15)$$

The n -th order correction to $F(t, x, y)$ is defined as

$$F_n(t, x, y) = -\frac{i\hbar}{2m} \frac{\partial}{\partial q} [(xy|e^{i\hat{L}_{pb}t}\mathcal{K}_n(t)|0q) - (xy|e^{i\hat{L}_{pb}t}\mathcal{K}_n(t)|q0)]_{q=0}. \quad (2.16)$$

Using the recursion relation given in Eq. (2.8), we find that the n -th order term is connected to the $(n-1)$ -th order term through the relation

$$F_n(t, x, y) = i \int_0^t dt_1 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dq_1 dq_2 (xy|e^{i\hat{L}_{pb}t}\tilde{L}_1(t_1) \times e^{-i\hat{L}_{pb}t_1}|q_1 q_2) F_{n-1}(t_1, q_1, q_2). \quad (2.17)$$

Noting that,

$$(q_3 q_4|\hat{L}_1|q_1 q_2) = \frac{1}{\hbar} \delta(q_1 - q_3) \delta(q_2 - q_4) \times [H_1(q_1) - H_1(q_2)], \quad (2.18)$$

we find the central recursion relation,

$$F_n(x, y, t) = -\frac{i}{\hbar} \int_{-\infty}^{\infty} dq_1 \int_{-\infty}^{\infty} dq_2 \int_0^t dt_1 \times (x, y|e^{i\hat{L}_{pb}(t-t_1)}|q_1, q_2) \times [H_1(q_2) - H_1(q_1)] F_{n-1}(q_1, q_2, t_1), \quad (2.19)$$

which allows one to find the n -th order correction for the time-evolved flux operator in terms of the $(n-1)$ -th order term.

This recursion relation may be solved since the supermatrix elements of the parabolic barrier Liouville operator are known analytically. Using the known configuration space matrix elements of the propagator for the parabolic barrier and inserting them into Eq. (2.14), one readily finds that the unperturbed supermatrix elements for the parabolic barrier Liouville operator are

$$(x_1 x_2|e^{i\hat{L}_{pb}t}|x_3 x_4) = \frac{m\omega^\ddagger}{2\pi\hbar \sinh(\omega^\ddagger t)} \times \exp\left[i \frac{m\omega^\ddagger}{2\hbar \sinh(\omega^\ddagger t)} (x_2^2 + x_4^2 - x_1^2 - x_3^2) \cosh(\omega^\ddagger t)\right] \times \exp\left[i \frac{2m\omega^\ddagger}{2\hbar \sinh(\omega^\ddagger t)} (x_1 x_3 - x_2 x_4)\right]. \quad (2.20)$$

To obtain the rate, we need the infinite positive time integral of the time-evolved flux operator,

$$P_n(x, y) = \int_0^\infty dt F_n(x, y, t). \quad (2.21)$$

Using the well known identity,

$$\int_0^\infty dt \int_0^t dt_1 f(t-t_1) g(t_1) = \int_0^\infty dt f(t) \int_0^\infty dt_1 g(t_1), \quad (2.22)$$

we can rewrite the recursion relation given in Eq. (2.19) as

$$P_n(x, y) = -\frac{i}{\hbar} \int_{-\infty}^{\infty} dq_1 \int_{-\infty}^{\infty} dq_2 \int_0^\infty dt (x, y_x|e^{i\hat{L}_{pb}t}|q_1, q_2) \times [H_1(q_2) - H_1(q_1)] P_{n-1}(q_1, q_2). \quad (2.23)$$

B. The recursion relation in phase space

The purpose of this subsection is to rewrite the recursion relation, Eq. (2.23), in terms of the Wigner distribution function,

$$P_n^W(p, q) = \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} d\xi e^{ip\xi/\hbar} P_n(q - \xi/2, q + \xi/2). \quad (2.24)$$

For this purpose we note the inverse Wigner transform,

$$P_n(x, y) = \int_{-\infty}^{\infty} dp e^{-ip(y-x)/\hbar} P_n^W[p, (x+y)/2]. \quad (2.25)$$

Denote

$$A(t) \equiv \frac{m\omega}{2\hbar \sinh(\omega t)}, \quad (2.26)$$

insert Eq. (2.24) into the right-hand side of Eq. (2.23), take the Wigner transform of Eq. (2.23), use the explicit known result for the supermatrix element of $e^{i\hat{L}_{pb}t}$ as given in Eq. (2.20), and note that the ξ integration is straightforward leading to a δ function, to obtain the intermediate result,

$$P_n^W(p, q) = -\frac{i}{\pi\hbar} \int_{-\infty}^{\infty} dp' \int_{-\infty}^{\infty} dq_1 \int_{-\infty}^{\infty} dq_2 \int_0^\infty dt [H_1(q_2) - H_1(q_1)] P_{n-1}^W[p', (q_1 + q_2)/2] e^{-ip'(q_2 - q_1)/\hbar} \times e^{iA(t)[(q_2^2 - q_1^2) \cosh(\omega t) + 2q(q_2 - q_1)]} \frac{1}{2\hbar} \times \left[\frac{p}{m\omega} \sinh(\omega t) + q \cosh(\omega t) - \frac{1}{2} (q_2 + q_1) \right]. \quad (2.27)$$

Change variables,

$$x = \frac{1}{2} (q_2 + q_1), \quad y = \frac{1}{\hbar} (q_2 - q_1), \quad (2.28)$$

integrate Eq. (2.27) over the variable x using the notation,

$$x(t) \equiv q \cosh(\omega t) + \frac{p}{m\omega} \sinh(\omega t), \quad (2.29)$$

and

$$p_x(t) \equiv p \cosh(\omega t) + m\omega q \sinh(\omega t), \quad (2.30)$$

to find the instructive result,

$$\begin{aligned}
P_n^W(p, q) = & -\frac{i}{2\pi\hbar} \int_{-\infty}^{\infty} dp' \int_{-\infty}^{\infty} dy \int_0^{\infty} dt e^{iy[p_x(t)-p']} \\
& \times P_{n-1}^W[p', x(t)] \left[H_1 \left[x(t) + \frac{\hbar y}{2} \right] \right. \\
& \left. - H_1 \left[x(t) - \frac{\hbar y}{2} \right] \right]. \quad (2.31)
\end{aligned}$$

The dependence of the n -th order term of the projection operator on $\hbar$ can come from only two sources. One is the nonlinearity term $[H_1(x(t) + \frac{1}{2}\hbar y) - H_1(x(t) - \frac{1}{2}\hbar y)]$ and the other, if there is any $\hbar$ dependence in the $(n-1)$ -th Wigner distribution $P_{n-1}^W[p', x(t)]$. To make further progress, we Taylor expand the nonlinearity term in a series in $\hbar$,

$$\begin{aligned}
& \frac{1}{\hbar} \left\{ H_1 \left[x(t) + \frac{1}{2} \hbar y \right] - H_1 \left[x(t) - \frac{1}{2} \hbar y \right] \right\} \\
& = \sum_{j=0}^{\infty} \frac{(\hbar/2)^{2j}}{(2j+1)!} y^{2j+1} \frac{d^{2j+1}}{dx(t)^{2j+1}} H_1[x(t)], \quad (2.32)
\end{aligned}$$

and integrate each term of the series over the y variable, using the identity,

$$\begin{aligned}
& -\frac{i}{2\pi} \int_{-\infty}^{\infty} dy y^{2j+1} e^{iy[p_x(t)-p']} \\
& = (-1)^j \frac{\partial^{2j+1}}{\partial p'^{2j+1}} \delta[p' - p_x(t)]. \quad (2.33)
\end{aligned}$$

Integrating over the variable p' in Eq. (2.31) gives a central result of this subsection,

$$\begin{aligned}
P_n^W(p, q) = & -\int_0^{\infty} dt \sum_{j=0}^{\infty} \frac{1}{(2j+1)!} \\
& \times \left(-\frac{\hbar^2}{4} \right)^j \left\{ \frac{d^{2j+1}}{dx(t)^{2j+1}} H_1[x(t)] \right\} \\
& \times \left\{ \frac{\partial^{2j+1}}{\partial p_x(t)^{2j+1}} P_{n-1}^W[p_x(t), x(t)] \right\}. \quad (2.34)
\end{aligned}$$

Since $P_0^W(p, q)$ is known for the parabolic barrier [cf. Eq. (1.8)], one can systematically obtain the higher order terms of the projection operator by successive application of the central recursion relation given in Eq. (2.34).

The recursion relation given in Eq. (2.34) leads to an expansion of the projection operator in terms of both the nonlinearity and $\hbar^2$, where the leading order term is the parabolic barrier projection operator. If one knows the exact classical projection operator, one may use the same recursion relation to obtain an improved expansion of the quantum projection operator in terms of $\hbar^2$ only. The exact quantum projection operator may be written as

$$P^W(p, q) = \sum_{n=0}^{\infty} \hbar^{2n} \Delta P_n^W(p, q). \quad (2.35)$$

Summing the left-hand side of the recursion relation from $n=1$ to ∞ , using the $\hbar^2$ expansion of Eq. (2.35), and equating powers of $\hbar^2$ on both sides, leads to an integral equation for the m -th contribution in the expansion,

$$\begin{aligned}
& \Delta P_m^W(p, q) - P_0^W(p, q) \delta_{0m} \\
& = -\sum_{j=0}^m \int_0^{\infty} dt \left(-\frac{1}{4} \right)^j \frac{1}{(2j+1)!} \left\{ \frac{d^{2j+1}}{dx(t)^{2j+1}} H_1[x(t)] \right\} \\
& \times \left\{ \frac{\partial^{2j+1}}{\partial p_x(t)^{2j+1}} \Delta P_{m-j}^W[p_x(t), x(t)] \right\}, \quad (2.36)
\end{aligned}$$

where δ_{0m} is the Kronecker δ symbol. We shall see in the next section that these integral equations are readily solved, and that the zeroth order contribution is the exact classical projection operator.

III. PERTURBATION EXPANSION FOR ONE DIMENSIONAL SYSTEMS

In this section we will show how the recursion relation may be solved to obtain the leading order terms in the expansion of the quantum projection operator. Solutions will be provided in Sec. III A for the expansion about the parabolic barrier solution. The expansion about the exact classical solution is given in Sec. III B.

A. Parabolic barrier expansion

The n -th order contribution to the projection operator $P_n^W(p, q)$ may be further subdivided into a series in $\hbar^2$ such that

$$P_n^W(p, q) = \sum_{j=0}^{\infty} \hbar^{2j} P_{nj}^W(p, q). \quad (3.1)$$

In this subsection we will derive the explicit solutions for the P_{nj}^W 's from the parabolic barrier solution for the projection operator $P_0^W(p, q) = P_{00}^W(p, q)$, which has already been given in Eq. (1.8). Note that for any $j \neq 0$, $P_{0j}^W(p, q) = 0$.

The recursion relations involve high order partial derivatives of the projection operator functions with respect to the momentum. Since the zeroth order function is a step function, these derivatives will be higher order derivatives of Dirac δ functions whose arguments have the general form $p - g(q)$. To manipulate the coefficients of these delta function derivatives, we note the following identities for any function $f(p, q)$ which multiplies a delta function of the form $\delta[p - g(q)]$,

$$\delta' f = f[g(q), q] \delta' - f'[g(q), q] \delta, \quad (3.2)$$

$$\delta'' f = f[g(q), q] \delta'' - 2 f'[g(q), q] \delta' + f''[g(q), q] \delta, \quad (3.3)$$

$$\begin{aligned}
\delta''' f = & f[g(q), q] \delta''' - 3 f'[g(q), q] \delta'' + 3 f''[g(q), q] \delta' \\
& - f'''[g(q), q] \delta, \quad (3.4)
\end{aligned}$$

where for the sake of brevity we suppressed the arguments of the δ functions. The primes denote partial differentiation with respect to the momentum, so that, for example, $f'[g(q), q] \equiv [\partial f(p, q) / \partial p]_{p=g(q)}$.

The classical recursion relation is obtained by taking only the leading order term in the expansion in $\hbar^2$ given in Eq. (2.34),

$$P_{n0}^W(p, q) = - \int_0^\infty dt \frac{dH_1[x(t)]}{dx(t)} \frac{\partial P_{n-1,0}^W[p_x(t), x(t)]}{\partial p_x(t)}. \quad (3.5)$$

Insertion of the zeroth order function into Eq. (3.5), noting that

$$p_x(t) + m\omega x(t) = (p + m\omega q)e^{\omega t}, \quad (3.6)$$

gives the result,

$$P_{10}^W(p, q) = - \frac{1}{2\pi\hbar} \delta(p + m\omega q) \int_0^\infty dt \frac{dH_1[x(t)]}{dx(t)} e^{-\omega t}. \quad (3.7)$$

Because of the δ function, $x(t) = qe^{-\omega t}$, so that one may change integration variables from t to $z = qe^{-\omega t}$ to find that

$$P_{10}^W(p, q) = - \frac{1}{2\pi\hbar} \delta(p + m\omega q) \frac{H_1(q)}{\omega q}. \quad (3.8)$$

To obtain $P_{20}^W(p, q)$, we insert Eq. (3.8) into Eq. (3.5) and note that [using Eq. (3.6)]

$$\frac{\partial P_1^W[p_x(t), x(t)]}{\partial p_x(t)} = e^{-2\omega t} \frac{\partial \delta(p + m\omega q)}{\partial p} \frac{H_1[x(t)]}{\omega x(t)}. \quad (3.9)$$

Using Eq. (3.2), noting that

$$\frac{\partial x(t)}{\partial p} = \frac{1}{m\omega} \sinh(\omega t), \quad (3.10)$$

and changing variables from t to $z = qe^{-\omega t}$, one finds the result,

$$P_{20}^W(p, q) = \frac{1}{2\pi\hbar} \frac{H_1(q)^2}{2\omega^2 q^2} \left[\frac{\partial \delta(p + m\omega q)}{\partial p} + \frac{1}{p} \delta(p + m\omega q) \right]. \quad (3.11)$$

The computation presented above is in fact a very complicated way of getting the Taylor expansion of the exact classical projection operator in terms of the nonlinearity H_1 . The one-dimensional exact classical projection operator is just the Heaviside function on the classical separatrix,

$$\Delta P_0^W(p, q) = \frac{1}{2\pi\hbar} h(p \mp [-2mV(q)]^{1/2}), \quad (3.12)$$

where the $-$ sign is to be taken for negative q and the $+$ sign for positive q . We will see in the next subsection that Eq. (3.12) is indeed a solution for the leading order term of the integral Eq. (2.36).

The next step is to find the explicit solution for the leading order term (P_{11}) in the expansion in $\hbar^2$. Since $P_{00}^W(p, q)$ is known, we find from Eq. (2.34), equating all terms of order $\hbar^2$, that the correction term is

$$2\pi\hbar P_{11}^W(p, q) = \frac{1}{24} \int_0^\infty dt H_1^{(3)}[x(t)] \times \delta''[p_x(t) + m\omega x(t)]. \quad (3.13)$$

This may be further simplified by virtue of Eq. (3.6) and by noting that $\delta''[(p + m\omega q)e^{\omega t}] = e^{-3\omega t} \delta''(p + m\omega q)$. Using Eqs. (3.2), (3.3), and (3.10), the fact that when $p = -m\omega q$, $x(t) = qe^{-\omega t}$, $p_x(t) = pe^{-\omega t}$, using the substitution $z = qe^{-\omega t}$, we find the result,

$$\begin{aligned} 2\pi\hbar P_{11}^W(p, q) &= g_{2(11)}(q) \delta''(p + m\omega q) + g_{1(11)}(q) \\ &\times \delta'(p + m\omega q) + g_{0(11)} \delta(p + m\omega q) \\ &= \frac{1}{24\omega} \left\{ \delta''(p + m\omega q) \int_0^q dz \frac{z^2}{q^3} H_1^{(3)}(z) \right. \\ &\quad - 2\delta'(p + m\omega q) \int_0^q dz \frac{z(q^2 - z^2)}{q^4} \\ &\quad \times H_1^{(4)}(z) \frac{1}{2m\omega} + \delta(p + m\omega q) \\ &\quad \times \left. \int_0^q dz \frac{(q^2 - z^2)^2}{q^5} H_1^{(5)}(z) \frac{1}{4m^2\omega^2} \right\}. \end{aligned} \quad (3.14)$$

One can carry out the remaining integrals in Eq. (3.14) by successive integrations by parts; the explicit results are

$$g_{2(11)}(q) = \frac{1}{24\omega} \left[\frac{H_1''(q)}{q} - \frac{2H_1'(q)}{q^2} + 2\frac{H_1(q)}{q^3} \right], \quad (3.15)$$

$$g_{1(11)}(q) = \frac{1}{24m\omega^2} \left[-\frac{2H_1''(q)}{q^2} + \frac{6H_1'(q)}{q^3} - \frac{6H_1(q)}{q^4} \right], \quad (3.16)$$

and

$$g_{0(11)}(q) = -\frac{1}{m\omega q} \left[g_{1(11)}(q) + \frac{H_1^{(4)}(0)}{96m\omega} \right]. \quad (3.17)$$

In this manner one can, in principle, continue to derive explicit expressions for the $P_{ij}^W(p, q)$'s. The computation becomes tedious, though, and one should resort to a symbolic computer language such as Mathematica or Maple to generate them.

B. Expansion in $\hbar^2$ about the classical limit

To obtain the expansion in $\hbar^2$ about the classical limit, one must solve Eq. (2.36) for successive increasing orders in $\hbar^2$. The zeroth order equation is

$$\begin{aligned} \Delta P_0^W(p, q) - P_0^W(p, q) &= - \int_0^\infty dt \left\{ \frac{d}{dx(t)} H_1[x(t)] \right\} \\ &\times \left\{ \frac{\partial}{\partial p_x(t)} \Delta P_0^W[p_x(t), x(t)] \right\}. \end{aligned} \quad (3.18)$$

It is instructive to show that the solution of this integral equation is a step function about the classical separatrix, as already given in Eq. (3.12), as the methodology will help us in solving for the higher order terms.

To proceed, we set up some notation. We let $p_{sx}^\pm$ denote the argument of the classical projection operator [cf. Eq. (3.12)], where the superscript $+$ ($-$) is for positive (negative) q ,

$$p_{sx}^\pm(t) \equiv p_x(t) \pm [-2mV(x(t))]^{1/2}. \quad (3.19)$$

At $t=0$, $x(0)=q$, $p_x(0)=p$, so that

$$p_{sx}^\pm(0) = p \pm [-2mV(q)]^{1/2}. \quad (3.20)$$

From Eqs. (2.29) and (2.30) it follows that

$$\dot{p}_{sx}^\pm(t) = m\omega^2 x(t) \mp \frac{p_x(t)V'[x(t)]}{[-2mV(x(t))]^{1/2}}, \quad (3.21)$$

where the dot denotes the time derivative.

Since $x(0)=q$ and $p_x(0)=p$, one may write for any $\Delta P_m^W(p, q)$,

$$\Delta P_m^W(p, q) = \lim_{t \rightarrow \infty} \Delta P_m^W[p_x(t), x(t)] - \int_0^\infty dt \frac{d\Delta P_m^W[p_x(t), x(t)]}{dt}. \quad (3.22)$$

In the limit that $t \rightarrow \infty$, the coordinate $x(t)$ and the momentum $p_x(t)$ grow indefinitely. Since the potential goes to a constant at large values of the coordinate, one notes that

$$\lim_{t \rightarrow \infty} p_{sx}(t) = (p + m\omega q)e^{\omega t}. \quad (3.23)$$

Equation (3.22) allows us to rewrite the classical integral Eq. (3.18) as

$$\begin{aligned} \lim_{t \rightarrow \infty} \Delta P_0^W[p_x(t), x(t)] - P_{00}^W(p, q) \\ = \int_0^\infty dt \left\{ \frac{d\Delta P_0^W[p_x(t), x(t)]}{dt} - H_1'[x(t)] \frac{\partial \Delta P_0^W[p_x(t), x(t)]}{\partial p_x(t)} \right\}. \end{aligned} \quad (3.24)$$

To show that the classical projection operator is a solution of this equation, we note that

$$\begin{aligned} \frac{\partial}{\partial p_x(t)} [\delta(p_x(t) \pm [-2mV(x(t))]^{1/2})] \\ = \frac{1}{\dot{p}_{sx}^\pm(t)} \frac{d}{dt} [\delta(p_{sx}^\pm(t))]. \end{aligned} \quad (3.25)$$

Furthermore, when $p_x(t)$ and $x(t)$ are on the separatrix, one readily finds that $\dot{p}_{sx}^\pm(t) = H_1'[x(t)]$. It is then obvious that if ΔP_0^W is the classical projection operator, then the integrand on the right-hand side of Eq. (3.24) is identically zero. But from Eq. (3.23) one notes that $\lim_{t \rightarrow \infty} h[p_{sx}(t)] = h[(p + m\omega q)e^{\omega t}] = h(p + m\omega q)$. Therefore, the left-hand side of Eq. (3.24) is also zero, proving that the leading order term in the expansion is indeed the classical projection operator.

Following the same logic, the integral Eq. (2.36) for the $\hbar^2$ expansion for any $m \neq 0$ can be rewritten as

$$\begin{aligned} 0 = \int_0^\infty dt \left\{ \frac{d\Delta P_m^W[p_x(t), x(t)]}{dt} - \sum_{j=0}^m \left(-\frac{1}{4} \right)^j \frac{1}{(2j+1)!} \frac{d^{2j+1} H_1[x(t)]}{dx(t)^{2j+1}} \right. \\ \left. \times \frac{\partial^{2j+1} \Delta P_{m-j}^W[p_x(t), x(t)]}{\partial p_x(t)^{2j+1}} \right\}, \end{aligned} \quad (3.26)$$

where we expressly assumed that $\lim_{t \rightarrow \infty} \Delta P_m^W[p_x(t), x(t)] = 0$. Noting that Eq. (3.26) is correct for any H_1 , we may conclude that the integrand of the equation must be identically zero. This then leads to the central differential form for the expansion

$$\begin{aligned} \frac{d\Delta P_m^W[p_x(t), x(t)]}{dt} \\ = \sum_{j=0}^m \left(-\frac{1}{4} \right)^j \frac{1}{(2j+1)!} \left\{ \frac{d^{2j+1}}{dx(t)^{2j+1}} H_1[x(t)] \right\} \\ \times \left\{ \frac{\partial^{2j+1}}{\partial p_x(t)^{2j+1}} \Delta P_{m-j}^W[p_x(t), x(t)] \right\}. \end{aligned} \quad (3.27)$$

The specific form of the differential equation for the first order term in the expansion is thus

$$\begin{aligned} \frac{d\Delta P_1^W[p_x(t), x(t)]}{dt} \\ = H_1'[x(t)] \frac{\partial \Delta P_1^W[p_x(t), x(t)]}{\partial p_x(t)} - \frac{1}{24} H_1^{(3)}[x(t)] \frac{\partial^3 \Delta P_0^W[p_x(t), x(t)]}{\partial p_x(t)^3}, \end{aligned} \quad (3.28)$$

where $H_1', H_1^{(3)}$ denote the first and third derivatives with respect to the argument.

Guided by the perturbation theory results of the previous subsection, we suggest that the solution of the integral equation takes the form

$$\begin{aligned} 2\pi\hbar \Delta P_1(p, \pm q) = g_{21}^\pm(q) \delta''[p_{sx}^\pm(0)] + g_{11}^\pm(q) \\ \times \delta'[p_{sx}^\pm(0)] + g_{01}^\pm(q) \delta[p_{sx}^\pm(0)], \end{aligned} \quad (3.29)$$

so that

$$\begin{aligned} \frac{d\Delta P_1^W[p_x(t), \pm x(t)]}{dt} = \dot{p}_{sx}^\pm(t) \frac{\partial \Delta P_1^W[p_x(t), \pm x(t)]}{\partial p_x(t)} \\ + \frac{1}{2\pi\hbar} \frac{p_x(t)}{m} \{ g_{01}^{\pm'}[x(t)] \delta[p_{sx}^\pm(t)] \\ + g_{11}^{\pm'}[x(t)] \delta'[p_{sx}^\pm(t)] \\ + g_{21}^{\pm'}[x(t)] \delta''[p_{sx}^\pm(t)] \}. \end{aligned} \quad (3.30)$$

Inserting this result into the differential Eq. (3.28), using the identities given in Eqs. (3.2)–(3.4), one finds that the differential equation, after rearranging, takes the form

$$F_{21}(\pm q)\delta''[p_{sx}^{\pm}(0)] + F_{11}(\pm q)\delta'[p_{sx}^{\pm}(0)] + F_{01}(\pm q) \\ \times \delta[p_{sx}^{\pm}(0)] = 0, \quad (3.31)$$

where

$$F_{21}(\pm q) = 3g_{21}^{\pm}(q) \frac{\pm V'(q)}{\sqrt{-2mV(q)}} \\ \mp g_{21}'^{\pm}(q) \frac{\sqrt{-2mV(q)}}{m} + \frac{H_1^{(3)}(q)}{24}, \quad (3.32)$$

$$F_{11}(\pm q) = 2g_{11}^{\pm}(q) \frac{\pm V'(q)}{\sqrt{-2mV(q)}} \mp g_{11}'^{\pm}(q) \\ \times \frac{\sqrt{-2mV(q)}}{m} - \frac{2}{m} g_{21}'^{\pm}(q), \quad (3.33)$$

$$F_{01}(\pm q) = g_{01}^{\pm}(q) \frac{\pm V'(q)}{\sqrt{-2mV(q)}} \mp g_{01}'^{\pm}(q) \\ \times \frac{\sqrt{-2mV(q)}}{m} - \frac{1}{m} g_{11}'^{\pm}(q). \quad (3.34)$$

Equating the coefficients F_{ji} to zero leads to three differential equations which are readily solved. The various constants of integration are determined by the condition that for a purely parabolic barrier, each of the functions must be identically zero and that the g_{ji} 's do not diverge at $q=0$. One then finds the explicit expressions for the first order correction

$$g_{21}^{\pm}(q) = \pm \frac{m}{24} \frac{1}{[-2mV(q)]^{1/2}} \left\{ V''(q) - \frac{[V'(q)]^2}{2V(q)} \right\}, \quad (3.35)$$

$$g_{11}^{\pm}(q) = \frac{1}{24} \frac{1}{V(q)} \left\{ V''(q) - \frac{3}{4} \frac{[V'(q)]^2}{V(q)} - \frac{1}{2} m\omega^2 \right\}, \quad (3.36)$$

$$g_{01}^{\pm}(q) = \mp \frac{1}{[-2mV(q)]^{1/2}} [g_{11}^{\pm}(q) - g_{11}^{\pm}(0)]. \quad (3.37)$$

One can readily see that these results reduce to Eqs. (3.15)–(3.17) when expanding to leading order in H_1 . Also, note that as $p, q \rightarrow \infty$ the solution for ΔP_1^W goes to zero, since the argument of the δ functions $p_{sx}^{\pm}(0) \rightarrow p \neq 0$.

IV. APPLICATION TO THE SYMMETRIC AND ASYMMETRIC ECKART BARRIER

A. Preliminaries

The exact quantum rate for a one-dimensional system may be written as²⁵

$$k(T) = Q_r(T)^{-1} \int_{-\infty}^{\infty} dp dq P^W(p, q) \rho_W[\hat{F}(\beta, q_{ds}); p, q]. \quad (4.1)$$

The rate may be symbolically expanded as

$$k(T) \equiv \sum_{j=0}^{\infty} \hbar^{2j} \Delta k_j(T) \\ = \frac{2\pi\hbar}{Q_r(T)} \int_{-\infty}^{\infty} dp \int_{-\infty}^{\infty} dq \rho_W[\hat{F}(\beta, q_{ds}); p, q] \\ \times \sum_{j=0}^{\infty} \hbar^{2j} \Delta P_j^W(p, q) \\ = \sum_{j=0}^{\infty} \sum_{l=0}^{\infty} \hbar^{2j} k_{jl}(T) \\ = \frac{2\pi\hbar}{Q_r(T)} \int_{-\infty}^{\infty} dp \int_{-\infty}^{\infty} dq \rho_W[\hat{F}(\beta, q_{ds}); p, q] \\ \times \sum_{j=0}^{\infty} \sum_{l=0}^{\infty} \hbar^{2j} \Delta P_{jl}^W(p, q). \quad (4.2)$$

With this notation, $\Delta k_j(T)$ and $k_{jl}(T)$ are partial rate constants (in units of $\hbar^{2j}$) associated with the j -th and jl -th contributions to the projection operator, respectively. The partial rates have all orders of $\hbar$ in them, since the flux Wigner distribution function is evaluated exactly. The notation of Eq. (4.2) serves just to remind us that we are using the appropriate expansion term of the projection operator.

The purpose of this subsection is to derive explicit expressions for $\Delta k_1(T)$ and $k_{10}(T)$ using the results of the previous section for $\Delta P_1(p, q)$ and $P_{10}(p, q)$, respectively. The explicit expression for the Wigner flux distribution function may be written as

$$\rho_W[\hat{F}(\beta, q_{ds}); p, q] \\ = \frac{1}{2m\pi} \int_0^{\infty} d\xi \sin\left(\frac{p\xi}{\hbar}\right) \left[\left\langle q - \frac{\xi}{2} \left| e^{-1/2\beta\hat{H}} \right| q_{ds} \right\rangle \right. \\ \times \left\langle \frac{\partial}{\partial q_{ds}} \left| e^{-1/2\beta\hat{H}} \right| q + \frac{\xi}{2} \right\rangle \\ \left. - \left\langle q - \frac{\xi}{2} \left| e^{-1/2\beta\hat{H}} \right| \frac{\partial}{\partial q_{ds}} \right\rangle \left\langle q_{ds} \left| e^{-1/2\beta\hat{H}} \right| q + \frac{\xi}{2} \right\rangle \right], \quad (4.3)$$

where the notation $\langle q | e^{-\beta\hat{H}/2} | (\partial/\partial q_{ds}) \rangle$ is shorthand notation for $(\partial/\partial q') \langle q | e^{-\beta\hat{H}/2} | q' \rangle |_{q'=q_{ds}}$.

The dependence of the flux density on the momentum p comes only through the $\sin(p\xi/\hbar)$ term in Eq. (4.3). We may thus integrate the momentum over the product of this sin function and the leading order correction to the projection operator. Using Eqs. (3.2)–(3.4), the notation,

$$p^{\pm}(q) \equiv \mp [-2mV(q)]^{1/2}, \quad (4.4)$$

and the fact that

$$\int_{-\infty}^{\infty} dp \delta'[p_{sx}^{\pm}(0)] = \int_{-\infty}^{\infty} dp \delta''[p_{sx}^{\pm}(0)] = 0, \quad (4.5)$$

one finds

TABLE I. Perturbation theory results for the symmetric Eckart potential.

u	$\tilde{k}_{00}$	$\tilde{k}_{10}$	$\tilde{k}_{11}$	$\tilde{k}_{00} + \tilde{k}_{10} + \tilde{k}_{11}$	$\Delta\tilde{k}_0$	$\Delta\tilde{k}_1$	$\Delta\tilde{k}_0 + \Delta\tilde{k}_1$	$\tilde{k}_{\text{ex}}$
2	1.23	0.04	-0.03	1.24	1.27	-0.04	1.23	1.22
4	2.09	0.08	-0.09	2.08	2.16	-0.10	2.06	2.07
6	5.74	0.01	-0.27	5.48	5.75	-0.32	5.43	5.20
8	29.3	-1.8	-1.3	26.2	26.4	-1.6	24.8	21.8
10	248	-5.3	-7.5	235	204	-9.8	194	162
12	3058	-48.2	-62.0	2948	2700	-107	2593	1970

$$\begin{aligned}
& \int_{-\infty}^{\infty} dp \Delta P_1(p, \pm q) \sin\left(\frac{p\xi}{\hbar}\right) \\
&= \left[g_{01}^{\pm}(q) - g_{21}^{\pm}(q) \frac{\xi^2}{\hbar^2} \right] \sin\left[\frac{p^{\pm}(q)\xi}{\hbar}\right] \\
&\quad - g_{11}^{\pm}(q) \frac{\xi}{\hbar} \cos\left[\frac{p^{\pm}(q)\xi}{\hbar}\right]. \quad (4.6)
\end{aligned}$$

This means that the quantum correction term may be written as

$$\begin{aligned}
& Q_r(T) \Delta k_1(T) \\
&= \int_{-\infty}^0 dq \left\{ \left[g_{01}^-(q) - g_{11}^-(q) \frac{\partial}{\partial p} + g_{21}^-(q) \frac{\partial^2}{\partial p^2} \right] \right. \\
&\quad \times \rho_W[\hat{F}(\beta, q_{ds}); p, q] \left. \right\} \Big|_{p=p^-(q)} \\
&\quad + \int_0^{\infty} dq \left\{ \left[g_{01}^+(q) - g_{11}^+(q) \frac{\partial}{\partial p} + g_{21}^+(q) \frac{\partial^2}{\partial p^2} \right] \right. \\
&\quad \times \rho_W[\hat{F}(\beta, q_{ds}); p, q] \left. \right\} \Big|_{p=p^+(q)}. \quad (4.7)
\end{aligned}$$

Similarly, one finds that

$$\begin{aligned}
& Q_r(T) k_{01}(T) \\
&= \int_{-\infty}^{\infty} dq g_{0(01)}(q) \rho_W[\hat{F}(\beta, q_{ds}); p = -m\omega q, q] \quad (4.8)
\end{aligned}$$

and

$$\begin{aligned}
& Q_r(T) k_{11}(T) \\
&= \int_{-\infty}^{\infty} dq \left\{ \left[g_{0(11)}(q) - g_{1(11)}(q) \frac{\partial}{\partial p} + g_{2(11)}(q) \frac{\partial^2}{\partial p^2} \right] \right. \\
&\quad \times \rho_W[\hat{F}(\beta, q_{ds}); p, q] \left. \right\} \Big|_{p=-m\omega q}. \quad (4.9)
\end{aligned}$$

Expressions for higher order terms in the expansion are readily obtained in the same manner, though computations become increasingly tedious.

B. The symmetric Eckart barrier

The symmetric Eckart potential has the form

$$V(q) = V^{\ddagger} \operatorname{sech}^2(aq/2). \quad (4.10)$$

Using reduced units,

$$x = \left(\frac{mV^{\ddagger}}{\hbar^2} \right)^{1/2} q, \quad \alpha = \frac{2\pi\sqrt{2mV^{\ddagger}}}{\hbar a} = \frac{2\pi V^{\ddagger}}{\hbar\omega^{\ddagger}}, \quad (4.11)$$

the tunneling coefficient Γ becomes a function of the reduced time α and the reduced temperature $u = \hbar\beta\omega^{\ddagger} = (2\pi/a)\beta V^{\ddagger}$. In order to compare with previous computations, all results presented in this subsection were calculated with $\alpha = 12$.

The exact energy dependent transmission coefficient for the symmetric Eckart potential is³¹

$$\Gamma(E) = \frac{\cosh(2\alpha\sqrt{E/V^{\ddagger}}) - 1}{\cosh(2\alpha\sqrt{E/V^{\ddagger}}) + \cosh(\sqrt{4\alpha^2 - \pi^2})}. \quad (4.12)$$

The thermal transmission coefficient is obtained by numerical integration using the canonical distribution,

$$\Gamma(\beta) = \beta \int_0^{\infty} dE \Gamma(E) e^{-\beta E}. \quad (4.13)$$

Computation of the numerically exact Wigner flux distribution was carried out with the Monte Carlo methodology described in Refs. 25 and 32. Results are presented in Table I. All transmission coefficients are denoted by a “tilde” in all tables to distinguish them from the rate. The transmission coefficients are defined as $\tilde{k} \equiv k/k_{\text{cl}}$, where the classical rate is $k_{\text{cl}} = (1/\hbar\beta) e^{-\beta V^{\ddagger}}$. Results are given for the exact quantum transmission coefficient, the expansion about the parabolic barrier estimate for the rate [$\tilde{k}_{\text{QTST}}(T) = \tilde{k}_{00}(T)$], and the expansion about the classical separatrix estimate for the rate [$\Delta\tilde{k}_0(T)$]. Interestingly, at high temperature, the QTST estimate (k_{00}) is lower than the mixed quantum-classical estimate ($\Delta\tilde{k}_0$). This can be understood by inspection of Fig. 1, where the temperature dependent Wigner flux distributions are plotted for various temperatures. Shown also in the figure are the parabolic barrier and the classical separatrices (the separatrix for positive q is reflected to the positive p plane, for the sake of clarity). Note that for high temperatures, there is only one positive peak for the flux density. The exact separatrix covers a broader region of the phase space than the parabolic barrier separatrix so that the estimate is larger. As the temperature is lowered, the single positive peak bifurcates and one gets a series of alternating positive and negative peaks, whose magnitude decreases with increasing momentum. At low temperatures, the broader classical separatrix receives a larger contribution from the negative peak, hence the estimate $\Delta\tilde{k}_0(T)$ is lower than $\tilde{k}_{00}(T)$.

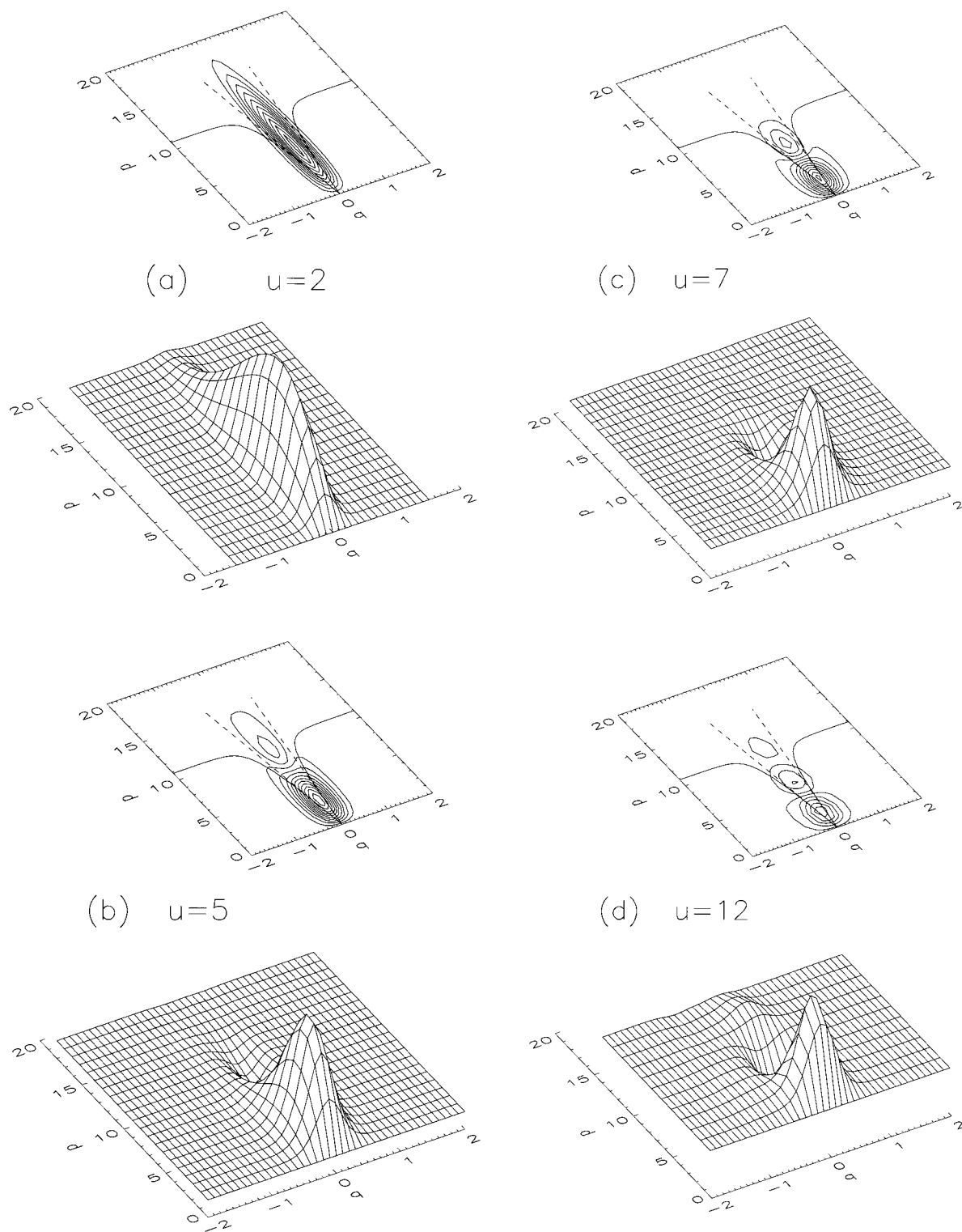


FIG. 1. Flux phase space distributions for the symmetric Eckart barrier. Panels (a)–(d) correspond to the temperatures $u = \hbar \beta \omega^\ddagger = 2, 5, 7$, and 12 , respectively. The solid and dashed lines superimposed on the contour plots are the classical and parabolic barrier separatrices, respectively (reflected about $p = 0$ for positive q). The dividing surface in all cases is at the barrier top ($q = 0$).

The temperature dependence of the flux distribution function may be understood from a semiclassical analysis and will be considered elsewhere.²⁹

For moderate to high temperature, $0 < u < 9$, the leading order classical anharmonic correction term $\tilde{k}_{10}$ is such that the sum of $\tilde{k}_{00} + \tilde{k}_{10}$ is practically identical to the result ob-

tained with the exact classical separatrix ($\Delta\tilde{k}_0$). This is encouraging as it implies that just the leading order classical correction term for the expansion about the parabolic barrier accounts for the classical nonlinearity. As discussed in the next section, for multidimensional systems one can compute the leading order correction terms to the parabolic barrier

estimate, but obtaining the leading order correction to the exact classical projection operator is difficult. The range of temperatures for which the parabolic barrier based estimate is accurate, is the usual range of temperatures in liquids.

One notes that the leading order quantum correction terms $\tilde{k}_{11}$, $\Delta\tilde{k}_1$ are always negative, thus bringing the respective rate estimates $(\tilde{k}_{00} + \tilde{k}_{10} + \tilde{k}_{11}, \Delta\tilde{k}_0 + \Delta\tilde{k}_1)$ substantially closer to the exact result $(\tilde{k}_{ex})$. Except for the lowest temperature ($u=12$), the two terms $\tilde{k}_{11}$ and $\Delta\tilde{k}_1$ are comparable in magnitude, again indicating that for most chemical applications, the QTST based perturbation theory is quite good. The correction terms increase in absolute magnitude as the temperature is lowered. It is evident, though, that one must include higher order terms in the expansion as the temperature is lowered, to account for the large extent of tunneling.

C. The asymmetric Eckart barrier

The asymmetric Eckart barrier has the form

$$V(q) = \frac{A}{1 + e^{-aq}} + \frac{B}{4 \cosh^2\left(\frac{1}{2}aq\right)}, \quad (4.14)$$

where $A = V_1 - V_2$, $B = (\sqrt{V_1} + \sqrt{V_2})^2$. As $q \rightarrow -\infty$, the potential $V(q) \rightarrow 0$. As $q \rightarrow \infty$ the potential tends to $V(q) \rightarrow V_1 - V_2$. The barrier height is $V^\ddagger \equiv V(q^\ddagger) - V(-\infty) = V_1$ and the barrier is located at $q^\ddagger = 1/a \ln[(B+A)/(B-A)]$. The tunneling coefficient is determined by the reduced temperature $u = \hbar\beta\omega^\ddagger$, where $\omega^\ddagger$ is the barrier frequency, and the two reduced times α_1 , α_2 are defined as

$$\alpha_i = \frac{2\pi V_i}{\hbar\omega^\ddagger}, \quad i=1,2. \quad (4.15)$$

Computations were performed for $\alpha_1 = 12$, $\alpha_2 = 48$. The dividing surface was chosen variationally, using the procedure described in Ref. 25. The variation of the zeroth order estimates for the rate as a function of the location of the dividing surface is shown for low temperature in Fig. 2. Note that within our numerical accuracy, this location is almost the same whether one chooses the parabolic barrier or the exact classical separatrix. The flux distributions for a few different locations of the dividing surface, at low temperature ($u=12$), are shown in Fig. 3. Note that at the optimized location, the flux distribution is very similar to the distribution in the symmetric case. As one moves away from the optimal location one obtains large contributions to the flux from the asymptotic region and the flux distribution becomes quite delocalized.

Numerical results for the transmission coefficients are given in Table II for the perturbation theory about the parabolic barrier and in Table III for the perturbation theory about the classical separatrix. We also provide in Table II our previous QTST estimates obtained by using the barrier top as the dividing surface ($\tilde{k}_{QTST}$). The variational optimization already provides a significant improvement for the rate estimate. As in the symmetric case, the mixed quantum-classical based results (Table III) are more accurate than those obtained by expansion about the QTST expression

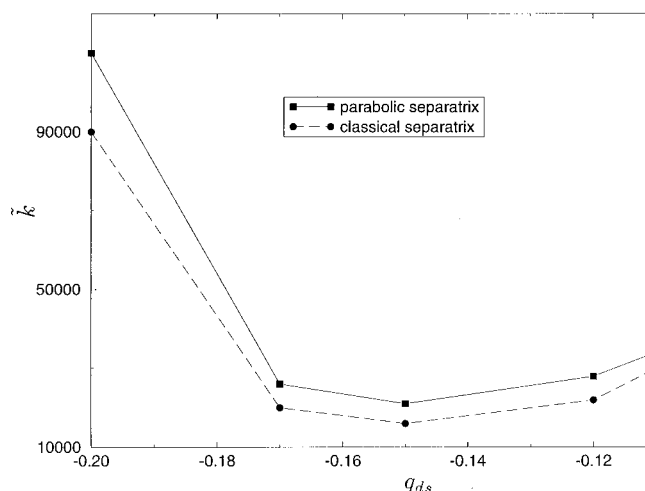


FIG. 2. Dependence of the transmission coefficient on the location of the dividing surface, for an asymmetric Eckart barrier. The barrier top is located at $q_{ds} = -0.12$. The dashed and solid lines are the transmission coefficients obtained by use of the classical and parabolic barrier projection operators, respectively.

(Table II). Differences, though, are not too big for moderate to high temperatures ($0 \leq u \leq 8$). For lower temperature, the deviation from the exact result is quite large; however, the combination of the variational choice of the dividing surface with the leading order perturbation theory correction term has reduced the error by a factor of 2.

V. MULTIDIMENSIONAL THEORY

The multidimensional Hamiltonian operator, with $N+1$ degrees of freedom, is assumed to be

$$\hat{H} = \frac{1}{2} \hat{p}_q^2 + \sum_j \frac{1}{2} \hat{p}_j^2 + V(\hat{q}, \hat{x}). \quad (5.1)$$

The coordinates q and x are taken already to be mass weighted: $q \rightarrow m^{1/2}q$, $x_j \rightarrow m_j^{1/2}x_j$, etc.³³ The vector x denotes N degrees of freedom; the special degree of freedom q will be defined shortly.

We assume that we are dealing with an activated reaction, that is, the potential surface V has a saddle point at the configuration space point $q^\ddagger, x^\ddagger$. Without loss of generality, the expansion of the potential about the saddle point is diagonal in the coordinates q, x such that

$$V(q, x) = V(q^\ddagger, x^\ddagger) - \frac{1}{2} \omega^{\ddagger 2} (q - q^\ddagger)^2, \\ + \frac{1}{2} \sum_{j=1}^N \omega_j^2 (x_j - x_j^\ddagger)^2 + H_1(q, x). \quad (5.2)$$

The coordinate q is thus identified as the unstable mode at the barrier, with barrier frequency $\omega^\ddagger$, the j -th coordinate x_j is a stable mode with frequency ω_j and Eq. (5.2) identifies H_1 as the nonlinear part of the potential surface. Without loss of generality, we will set the origin of the coordinate system at the barrier.

The flux operator at any point on the linear dividing surface q_{ds}, x takes the form

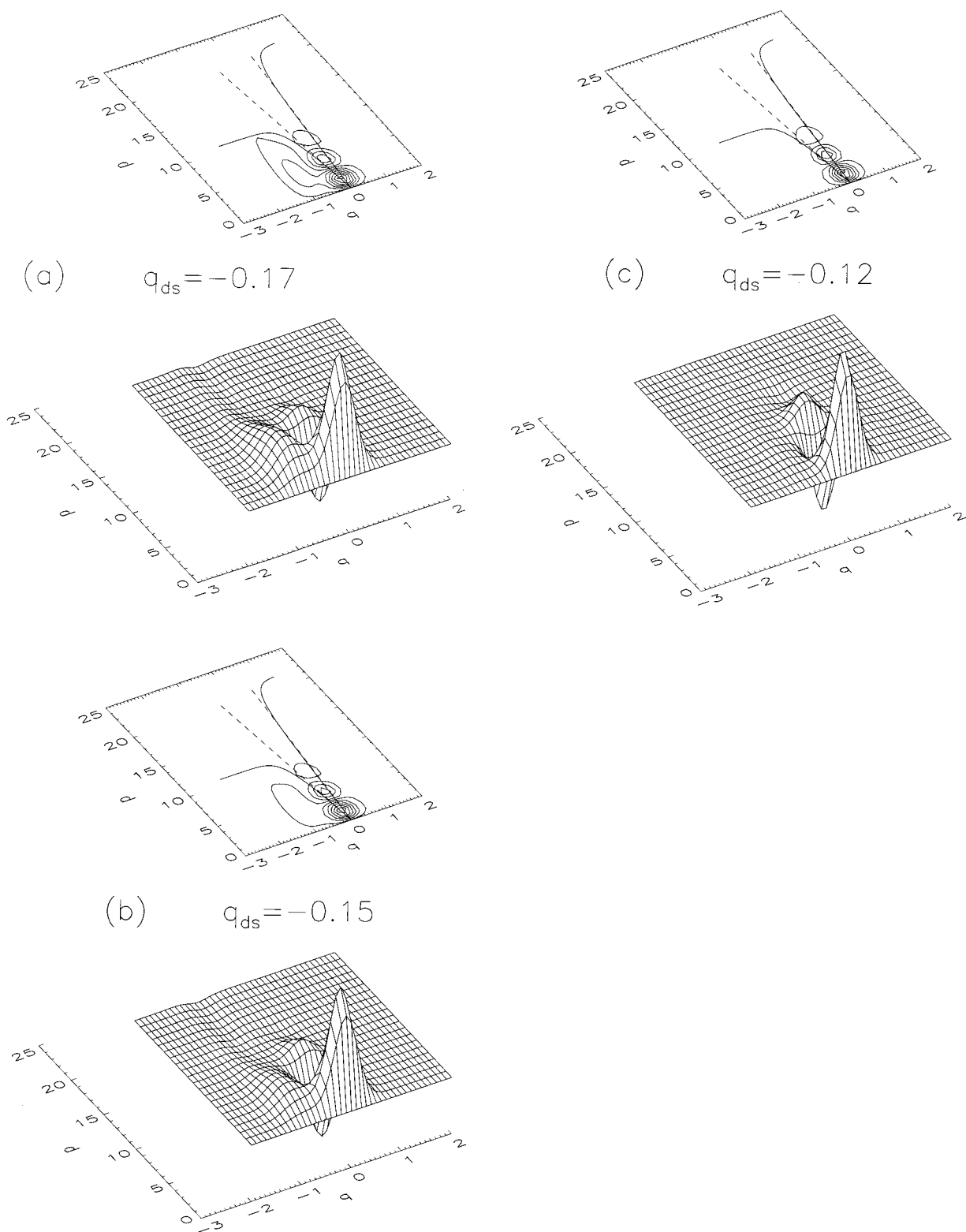


FIG. 3. Dependence of the flux phase space distributions for the asymmetric Eckart barrier on the dividing surface. The barrier top is located at $q_{ds} = -0.12$. The three panels are at the low temperature $u = 12$. The solid and dashed lines superimposed on the contour plots are the classical and parabolic barrier separatrices, respectively (reflected about $p = 0$ for positive q). The optimal dividing surface is at $q_{ds} \approx -0.15$ (cf. Tables II and III).

$$\hat{F}(q_{ds}, \underline{x}) = \frac{1}{2m} \delta(\hat{x} - \underline{x}) [\hat{p}_q \delta(\hat{q} - q_{ds}) + \delta(\hat{q} - q_{ds}) \hat{p}_q]. \quad (5.3)$$

The notation $\delta(\hat{x} - \underline{x})$ is a shorthand notation for a product of Dirac ' δ ' functions for each of the coordinates x_j . The thermal flux operator is defined as in Eq. (1.3),

$$\hat{F}(\beta, q_{ds}, \underline{x}) = e^{-\beta \hat{H}/2} \hat{F}(q_{ds}, \underline{x}) e^{-\beta \hat{H}/2}. \quad (5.4)$$

The multidimensional flux-side correlation function $C_{fs}(t)$ [cf. Eq. (1.2)] takes the form

$$C_{fs}(t) = \int_{-\infty}^{\infty} d\underline{x} \text{Tr} \hat{F}(\beta, q_{ds}, \underline{x}) e^{i\hat{H}t/\hbar} \hat{h}(\hat{q}) e^{-i\hat{H}t/\hbar}, \quad (5.5)$$

TABLE II. Variational quantum transition state theory and perturbation expansion results for an asymmetric Eckart potential.

u	q_{ds}	$\tilde{k}_{\text{QTST}}$	$\tilde{k}_0$	$\tilde{k}_{10}$	$\tilde{k}_{11}$	$\tilde{k}_{00} + \tilde{k}_{10} + \tilde{k}_{11}$	$\tilde{k}_{\text{ex}}$
2	-0.120	1.2	1.2	0.10	-0.07	1.2	1.2
4	-0.120	2.0	2.0	0.09	-0.09	2.0	2.0
6	-0.125	5.6	5.5	0.01	-0.21	5.3	5.3
8	-0.134	44	40	-4.3	-2.5	33.2	26
10	-0.145	1100	850	-87	-61	702	250
12	-0.155	28 000	21 000	-3800	-2000	15 200	4100

and the rate is as in Eq. (1.1). The multidimensional projection operator is thus formally defined exactly as in the one-dimensional case, cf. Eq. (1.5).

In this section we will provide the expansion for the projection operator in terms of both the nonlinearity H_1 and $\hbar^2$ about the parabolic barrier projection operator P_{00} which retains the same form as in the one-dimensional case,

$$P_{00}^W(p, q) = \frac{1}{2\pi\hbar} h(p + \omega^\dagger q). \quad (5.6)$$

The multidimensional Wigner distribution function for the flux operator is

$$\begin{aligned} \rho_W(\hat{F}(\beta, q_{ds}, \underline{x}); p, q, \underline{x}) \\ = \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} d\xi e^{i\xi p/\hbar} \langle q - \xi/2, \underline{x} | \hat{F}(\beta, q_{ds}) | q + \xi/2, \underline{x} \rangle. \end{aligned} \quad (5.7)$$

The rate is the multidimensional phase space trace of the product of the Wigner flux distribution and the projection operator.

The perturbation expansion for the projection operator follows closely the derivation as in the one-dimensional case. One notes that Eqs. (2.1)–(2.10) are general and applicable to systems with an arbitrary number of degrees of freedom. The differences come when one considers the representations of the supermatrix elements. For example, in the multidimensional case, Eqs. (2.12) and (2.14) take the form,

$$\begin{aligned} ((q_1, \underline{x}_1), (q_2, \underline{x}_2) | \hat{L} | (q_3, \underline{x}_3), (q_4, \underline{x}_4)) \\ = \frac{1}{\hbar} [\delta(q_2 - q_4) \delta(\underline{x}_2 - \underline{x}_4) \langle q_1, \underline{x}_1 | \hat{H} | q_3, \underline{x}_3 \rangle \\ - \delta(q_1 - q_3) \delta(\underline{x}_1 - \underline{x}_3) \langle q_4, \underline{x}_4 | \hat{H} | q_2, \underline{x}_2 \rangle], \end{aligned} \quad (5.8)$$

TABLE III. Variational mixed quantum-classical theory and perturbation expansion results for an asymmetric Eckart potential.

u	q_{ds}	$\Delta\tilde{k}_0$	$\Delta\tilde{k}_1$	$\Delta\tilde{k}_0 + \Delta\tilde{k}_1$	$\tilde{k}_{\text{ex}}$
2	-0.120	1.3	-0.08	1.2	1.2
4	-0.122	2.1	-0.10	2.0	2.0
6	-0.128	5.5	-0.23	5.4	5.3
8	-0.135	34	-3.2	30.8	26
10	-0.147	740	-85	655	250
12	-0.150	16 000	-2900	13 100	4100

$$\begin{aligned} ((q_1, \underline{x}_1), (q_2, \underline{x}_2) | e^{i\hat{L}t/\hbar} | (q_3, \underline{x}_3), (q_4, \underline{x}_4)) \\ = \langle q_1, \underline{x}_1 | e^{i\hat{H}t/\hbar} | q_3, \underline{x}_3 \rangle \langle q_4, \underline{x}_4 | e^{-i\hat{H}t/\hbar} | q_2, \underline{x}_2 \rangle. \end{aligned} \quad (5.9)$$

It follows that the multidimensional form of the recursion relation in configuration space, given in Eq. (2.23), takes the form,

$$\begin{aligned} P_n[(q, \underline{x}), (q', \underline{y}), t] \\ = -\frac{i}{\hbar} \int_{-\infty}^{\infty} dq_1 \prod_{j=1}^N dx_{j1} dq_2 \prod_{j=2}^N dx_{j2} \int_0^t dt_1 P_{n-1} \\ \times [(q_1, \underline{x}_1), (q_2, \underline{x}_2), t_1] ((q, \underline{x}), (q', \underline{y}) | e^{i\hat{L}_{pb}(t-t_1)} | \\ \times (q_1, \underline{x}_1), (q_2, \underline{x}_2)) [H_1(q_2, \underline{x}_2) - H_1(q_1, \underline{x}_1)]. \end{aligned} \quad (5.10)$$

The supermatrix elements of the parabolic barrier Liouville operator are known analytically,

$$\begin{aligned} ((q, \underline{x}), (q', \underline{y}) | e^{i\hat{L}_{pb}(t-t_1)} | (q_1, \underline{x}_1), (q_2, \underline{x}_2)) \\ = \frac{\omega^\dagger}{2\pi\hbar \sinh(\omega^\dagger t)} \exp \left[i \frac{\omega^\dagger}{2\hbar \sinh(\omega^\dagger t)} \right. \\ \times ((q'^2 + q_2^2 - q^2 - q_1^2) \cosh(\omega^\dagger t) + 2(qq_1 - q'q_2)) \left. \right] \\ \times \prod_{j=1}^N \frac{A_j(t)}{\pi} \exp[iA_j(t)[(y_j^2 + x_{j2}^2 - x_j^2 - x_{j1}^2) \\ \times \cos(\omega_j t) + 2(x_j x_{j1} - y x_{j2})], \end{aligned} \quad (5.11)$$

where the time-dependent coefficients $A_j(t)$, for $j=1, \dots, N$ are defined as

$$A_j(t) \equiv \frac{\omega_j}{2\hbar \sin(\omega_j t)}. \quad (5.12)$$

The next step is to rewrite the multidimensional recursion relation in phase space, using the Wigner transforms as in Eqs. (2.24) and (2.25). The resulting recursion relation takes the form,

$$\begin{aligned} P_n^W[(p, q), (p_x, \underline{x})] \\ = -\frac{i}{\hbar (2\pi)^{N+1}} \int_{-\infty}^{\infty} dp' dp'_x dq' d\underline{x}' \int_0^{\infty} dt \\ \cdot \exp\{i q' [p''(t) - p']\} \exp \left\{ \sum_{j=1}^N i x'_j [p''_{x_j}(t) - p'_{x_j}] \right\} \\ \times P_{n-1}^W[(p', q''(t)), (p'_x, \underline{x}''(t))] \cdot \left\{ H_1 \left[\left(q''(t) + \frac{\hbar q'}{2} \right), \right. \right. \\ \left. \left(\underline{x}''(t) + \frac{\hbar \underline{x}'}{2} \right) \right] - H_1 \left[\left(q''(t) - \frac{\hbar q'}{2} \right), \left(\underline{x}''(t) - \frac{\hbar \underline{x}'}{2} \right) \right] \right\}, \end{aligned} \quad (5.13)$$

where

$$p''(t) = p \cosh(\omega^\dagger t) + \omega^\dagger q \sinh(\omega^\dagger t), \quad (5.14)$$

$$q''(t) = q \cosh(\omega^\dagger t) + \frac{p}{\omega^\dagger} \sinh(\omega^\dagger t), \quad (5.15)$$

$$p_{x_j}''(t) = p_{x_j} \cos(\omega_j t) + \omega_j x_j \sin(\omega_j t), \quad j = 1, \dots, N, \quad (5.16)$$

$$x_j''(t) = x_j \cos(\omega_j t) + \frac{p_{x_j}}{\omega_j} \sin(\omega_j t), \quad j = 1, \dots, N. \quad (5.17)$$

As in the one-dimensional case, one may now expand the nonlinearity term as a series in $\hbar^2$ and then integrate term by term over the variables q', x' . Each such integration leads to derivatives of Dirac δ functions as in Eq. (2.33). Further integration over the variables p', p_x' leads in principle to the final form of the recursion relation. The resulting expression is lengthy; the higher order terms are in general quite difficult to compute. Below we detail the recursion relations for the leading order terms only; these remain relatively simple.

The n -th order term in the expansion in the nonlinearity is further expanded into a series in $\hbar^2$ as in Eq. (3.1). The leading order term in this expansion is the parabolic barrier projection operator P_{00} , which is known [cf. Eq. (5.6)], and is independent of the coordinates and momenta of the stable oscillators. The resulting classical recursion relation for the n -th order classical term P_{n0} is found from Eq. (5.13) to be

$$P_{n0}^W[(p, q), (p_x, x)] = - \int_0^\infty dt \sum_{j=0}^N \frac{\partial}{\partial p_{x_j}(t)} P_{(n-1)0}^W[(p''(t), q''(t)), \times (p_x''(t), x''(t))] \frac{\partial}{\partial x_j(t)} H_1[q''(t), x''(t)], \quad (5.18)$$

where, in the summation, we used the notation $x_0(t) = q''(t), p_{x_0}(t) = p''(t)$ for the index $j=0$. Inserting the known result for the parabolic barrier projection operator gives the leading order classical term,

$$P_{10}^W[(p, q), (p_x, x)] = - \frac{1}{2\pi\hbar} \delta(p + \omega^\dagger q) \int_0^\infty dt e^{-\omega^\dagger t} \times \frac{\partial}{\partial q''(t)} H_1[q''(t), x''(t)]. \quad (5.19)$$

Higher order terms become more complex, since now the first order correction P_{10} depends on all coordinates and momenta.

The leading order correction term in $\hbar^2$ is

$$2\pi\hbar P_{11}^W[(p, q), (p_x, x)] = \frac{1}{24} \int_0^\infty dt \frac{\partial^3}{\partial q''(t)^3} H_1[q''(t), x''(t)] \times \frac{\partial^2}{\partial p''(t)^2} \delta[p''(t) + \omega^\dagger q''(t)]. \quad (5.20)$$

Further simplification of this expression follows as in Eqs. (3.14) and (3.15)–(3.17).

Equations (5.19) and (5.20) are the central results of this section. Insertion of the resulting correction terms to the parabolic barrier projection operator into the rate expression allows the systematic evaluation of correction terms to the QTST rate estimate. The computation is not trivial; one will

need up to second order partial derivatives of the multidimensional Wigner flux distribution. However, simplifications will occur, for example, when treating dissipative systems described classically by generalized Langevin equations.

VI. DISCUSSION

A perturbation expansion has been presented for the quantum mechanical operator which projects onto the product space in reactive scattering. Two expansions were presented; an expansion for which the parabolic barrier limit is the leading order term and an expansion for which the classical limit of the projection operator is the leading order term. The various expressions were then combined with the exact quantum mechanical symmetrized thermal flux distribution to obtain quantum estimates for the thermal reaction rate. Both methods gave results of similar quality: the expansion about the classical separatrix converges faster, but the differences were not very great.

The parabolic barrier expansion has the immense advantage that it is relatively easy to compute also in multidimensional systems. In our one-dimensional computations we found that this approximation is quite accurate for temperatures above the crossover temperature ($u=2\pi$) between thermal activation and tunneling. This indicates that quantum transition state theory,²⁵ which is based on the parabolic barrier approximation, should be a viable theory for reactions in liquids, where usually, the temperature is above the crossover temperature.

The leading order correction terms involve higher order derivatives of the Dirac delta function about the separatrix. Guided by semiclassical considerations,²⁹ following some old work of Heller,³⁴ one may resum these terms to obtain a uniform Airy representation of the projection operator. This uniform approximation will be considered in some detail in Ref. 29.

This paper presented a first application of the variational aspect of quantum transition state theory. By varying the location of the dividing surface, looking for the minimal quantum flux, we found a significant improvement in the quality of the QTST prediction in asymmetric systems. The fact that a local flux minimum could be found is not trivial. At this point, we do not really understand why the QTST bounds the exact quantum result from above, nor why there should at all be a minimum in the flux as one varies the dividing surface. Of course, at high temperatures, one expects this behavior, based purely on classical mechanics considerations. But we found a variational solution also in the deep tunneling region, where the quantum transmission factor is three orders of magnitude greater than the classical.

Another new aspect of this paper is the phase space structure of the symmetrized thermal flux operator. At temperatures above crossover, the distribution is a Gaussian in the coordinate and a skewed Gaussian in the momentum. But as the temperature is lowered, the single Gaussian bifurcates into several peaks of alternating signs. This structure can be understood in terms of the semiclassical limit of the flux distribution and will be discussed in detail elsewhere.²⁹

In principle, one may use the methodology presented in this paper to evaluate all the higher order terms. Increasing

the order increases the order of the derivative of the Dirac delta function about the separatrix. Computation of higher order contributions to the rate involves then the computation of increasingly higher order derivatives of the quantum thermal flux in phase space. This becomes increasingly difficult. It is, though, of interest to understand the convergence properties of the expansion and to see whether one can indeed converge onto the exact quantum result simply by computing higher order terms.

The multidimensional perturbation expansion has also been considered. It seems to be routinely possible to evaluate the leading order term in the series about the quantum transition state theory expression. Higher order contributions, or a perturbation expansion about the classical limit of the projection operator, is probably prohibitive, since it is very often difficult to find the exact classical separatrix in multidimensional systems. It is, though, routinely possible to obtain the classical projection operator by running classical trajectories. The mixed quantum-classical theory outlined in Ref. 25, in which one substitutes the quantum projection operator by its classical limit, is amenable to numerical computation. One must compute at any point in phase space the exact quantum thermal flux, then one checks to see whether the given point is reactive or not. This is a well defined approximation, for which one could at least in principle also compute the next order correction term.

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